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200 Days Blog Posting Challenge

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> **Period:** *May 2024 → Sep 2025 (16 Months)*
> **Total:** 208 Days
> **Focus:** *STATistics → Machine Learning → Deep Learning → MLOps → Integration*
> **Link:** https://leahnote01.github.io/blog/

📘 Overview by Phase

Phase	Days	Theme	Focus	Key Outcome
🧱 **Phase I**	1–25	Foundations	Math, Probability, Inference	Built statistical and linear algebra foundation
🤖 **Phase II**	26–63	Core ML	Classification, Ensemble Methods	Developed core ML modeling skills
🧠 **Phase III**	64–108	Deep Learning	Transformers, Model Calibration	Connected DL theory with applied methods
📊 **Phase IV**	109–154	Applied Statistics	Regression, Unsupervised Learning	Revisited classical statistical ML
⚙️ **Phase V**	155–203	MLOps Systems	Data, Deployment, Monitoring	Designed scalable ML pipelines
💡 **Phase VI**	204–208	Integration & Review	Coding Practice, DL Review	Consolidated skills across domains

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🎯 Summary Dashboard

Category	Focus	Keywords
🟦 Statistics	Probability, Regression, Hypothesis Testing	CLT, Confidence Interval, ANOVA, Regression Diagnostics, Correlation
🟩 Machine Learning	Model Evaluation, Feature Engineering	PCA, LDA, SVM, Random Forest, XGBoost, Cross-Validation
🟨 Deep Learning	Transformers, Fine-Tuning, Evaluation	Self-Attention, LoRA, LLaMA, Conformal Prediction, OOD Detection
🟥 MLOps	System Design, Deployment, CI/CD	Dataflow, Drift Detection, Dockerization, Azure Integration, Monitoring

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🧩 Detailed Log (2024 – 2025)

📘 2024 — Building the Foundations of ML & DL

- ***From theory to intuition**: Focused on strengthening the mathematical and algorithmic backbone of machine learning.*
- *Revisited to connect statistical reasoning with core ML models and explored the foundations of deep learning architectures.*

🧱 Phase I – Foundations (Days 1–25)

> ***Goal:** Strengthen mathematical intuition and statistical fundamentals*

Key Topics:

- Linear Algebra (Matrix operations, Eigen decomposition)

- Probability Distributions (Joint, Marginal, Conditional)
- Hypothesis Testing (Z, t, Chi-square, Confidence Intervals)

Day	Title / Topic	Date	Category	Key Concepts
1	Basic Mathematics Review (1)	May 1 2024	🟦 STAT	Scalar, Norm, Matrix (Inverse & Rank)
2	Basic Mathematics Review (2)	May 15 2024	🟦 STAT	Norm (2), Eigenvectors & Eigenvalues
3	Basic Mathematics Review (3)	May 16 2024	🟦 STAT	Eigendecomposition & Symmetric Matrix
4	Basic Mathematics Review (4)	May 17 2024	🟦 STAT	Bayesian Statistics & Var-Cov Matrix
5	Principal Component Analysis (1)	May 20 2024	🟩 ML	PCA Definition & Algorithm Steps
6	Principal Component Analysis (2)	May 21 2024	🟩 ML	PCA Applications in ML & Further Explanation
7	Linear Regression (1)	May 22 2024	🟩 ML	Linear Regression Concept & Equation
8	Linear Regression (2)	May 23 2024	🟩 ML	Cost Function & Mean Squared Error
9	Linear Regression (3)	May 24 2024	🟩 ML	SSE & Model Evaluation
10	Gradient in Machine Learning	May 25 2024	🟩 ML	Gradient Concept & Optimization Basics
11	Gradient Descent (1)	May 28 2024	🟩 ML	Basic Steps & Learning Rate
12	Gradient Descent (2)	May 31 2024	🟩 ML	Mathematical Derivation of GD
13	Gradient Descent (3)	Jun 2 2024	🟩 ML	Types of GD (Batch, Stochastic, Mini-Batch)
14	R-Squared (1)	Jun 3 2024	🟩 ML	R^2 Concept & Formula
15	R-Squared (2)	Jun 4 2024	🟩 ML	Interpreting R^2 as Performance Metric
16	R-Squared (3)	Jun 5 2024	🟩 ML	Advanced Analysis on R^2
17	Regularization in ML	Jun 6 2024	🟩 ML	L1 & L2 Regularization, Bias-Variance Tradeoff
18	Principal Component Analysis (3)	Jun 8 2024	🟩 ML	Applying PCA & Scree Plot
19	Probability Review (1)	Jun 10 2024	🟦 STAT	Counting Principles & Random Variables
20	Probability Review (2)	Jun 11 2024	🟦 STAT	Joint, Marginal & Conditional Distributions
21	Probability Review (3)	Jun 14 2024	🟦 STAT	Continuous Distributions & Markov Chains
22	Statistics Review (1)	Jun 17 2024	🟦 STAT	Random Variable Properties
23	Statistics Review (2)	Jun 18 2024	🟦 STAT	LLN, CLT & Intro to Hypothesis Testing
24	Statistics Review (3)	Jun 19 2024	🟦 STAT	Z-Test, t-Test, Chi-Square Test
25	Statistics Review (4)	Jun 20 2024	🟦 STAT	p-Values, Confidence Intervals, Type I & II Errors

🤖 Phase II – Core Machine Learning (Days 26–63)

> *Goal: Build strong ML foundations through algorithm intuition and practical implementation*

Key Topics:

- Linear & Logistic Regression
- Perceptron, SVM, Decision Tree, Random Forest, XGBoost
- Cross Validation, Grid Search, F1 Score, AUC

- Bias-Variance Tradeoff & Model Selection

Day	Title / Topic	Date	Category	Key Concepts
26	Perceptron (1)	Jul 16 2024	🟩 ML	Single-Layer Perceptron & Training Process
27	Perceptron (2)	Jul 17 2024	🟩 ML	Training Steps & Decision Boundaries
28	Perceptron (3)	Jul 18 2024	🟩 ML	Convergence and Weight Update
29	Logistic Regression (1)	Jul 19 2024	🟩 ML	Sigmoid Function & Probabilistic Interpretation
30	Logistic Regression (2)	Jul 20 2024	🟩 ML	Cost Function & Cross-Entropy Loss
31	Logistic Regression (3)	Jul 22 2024	🟩 ML	Training in Scikit-Learn & Regularization
32	Support Vector Machine (1)	Jul 23 2024	🟩 ML	Hyperplanes & Mathematical Formulation
33	Support Vector Machine (2)	Jul 24 2024	🟩 ML	Non-Linear Cases & Kernel Concepts
34	Support Vector Machine (3)	Jul 25 2024	🟩 ML	Kernel SVM & Decision Boundaries
35	Decision Tree (1)	Jul 28 2024	🟩 ML	Gini Impurity & Information Gain
36	Decision Tree (2)	Jul 29 2024	🟩 ML	Entropy & Classification Error
37	Decision Tree (3) + Random Forest (1)	Jul 30 2024	🟩 ML	Tree Building & Ensemble Concepts
38	Random Forest (2)	Jul 31 2024	🟩 ML	Implementation & Bagging Principle
39	Linear Algebra & Matrix Review (KR)	Jul 31 2024	🟦 STAT	Matrix Operations & Vector Space
40	Statistics & Probability Lecture Review	Aug 1 2024	🟦 STAT	Comprehensive Statistics & Probability Summary
41	K-Nearest Neighbors (1)	Aug 1 2024	🟩 ML	Basic Concepts & Parametric vs Non-Parametric Models
42	K-Nearest Neighbors (2)	Aug 2 2024	🟩 ML	Distance Metrics (Euclidean, Manhattan, Minkowski, Cosine)
43	K-Nearest Neighbors (3)	Aug 5 2024	🟩 ML	Curse of Dimensionality
44	Data Preprocessing (1)	Aug 6 2024	🟩 ML	Handling Missing Data – Elimination & Imputation
45	Data Preprocessing (2)	Aug 7 2024	🟩 ML	Handling Categorical Data – Ordinal & One-Hot Encoding
46	Data Preprocessing (3)	Aug 8 2024	🟩 ML	Train/Test Split, Feature Scaling & Selection
47	Dimensionality Reduction (1)	Aug 12 2024	🟩 ML	PCA Overview & Linear Compression
48	Dimensionality Reduction (2)	Aug 13 2024	🟩 ML	Linear Discriminant Analysis (LDA)
49	Dimensionality Reduction (3)	Aug 14 2024	🟩 ML	Kernel PCA – Nonlinear Mapping
50	Dimensionality Reduction (4)	Aug 15 2024	🟩 ML	Implementing Kernel PCA in Python
51	Dimensionality Reduction (5)	Aug 17 2024	🟩 ML	Applying Kernel PCA to New Data
52	Pipeline in ML	Aug 18 2024	🟩 ML	Scikit-Learn Pipeline Concept & Example
53	Cross Validation (1)	Aug 19 2024	🟩 ML	Model Selection & K-Fold Cross Validation
54	Cross Validation (2)	Aug 20 2024	🟩 ML	Bias-Variance Tradeoff & Learning Curves
55	Cross Validation (3)	Aug 21 2024	🟩 ML	Grid Search & Hyperparameter Tuning
56	Cross Validation (4)	Aug 22 2024	🟩 ML	Confusion Matrix & F1 Score
57	Cross Validation (5)	Aug 23 2024	🟩 ML	ROC Curve & AUC Evaluation

Day	Title / Topic	Date	Category	Key Concepts
58	Class Imbalance Handling	Aug 25 2024	🟩 ML	Class Weights, Oversampling & Alternative Metrics
59	Ensemble Methods (1)	Aug 26 2024	🟩 ML	Bagging & Boosting Concepts
60	Ensemble Methods (2)	Aug 27 2024	🟩 ML	Majority Voting & Combining Classifiers
61	Ensemble Methods (3)	Aug 28 2024	🟩 ML	Majority Voting & Ensemble Evaluation
62	Ensemble Methods (4)	Aug 29 2024	🟩 ML	Bagging & Boosting Theory & Code
63	Ensemble Methods (5)	Aug 30 2024	🟩 ML	Random Forest & XGBoost Review

🧠 Phase III – Deep Learning (Days 64–108)

> *Goal: Understand neural architectures and large language models; bridge theory with reliability*

Key Topics:

- Neural Networks (MLP, CNN, RNN, Transformer)
- Attention Mechanisms & Positional Encoding
- LoRA, Prompt Engineering, KV Cache
- Calibration, OOD Detection, Conformal Prediction

Day	Title / Topic	Date	Category	Key Concepts
64	Deep Learning Lecture Review (1)	Sep 3 2024	🟨 DL	Basic Math, Supervised ML, MLP Overview
65	Deep Learning Lecture Review (2–3)	Sep 4 2024	🟨 DL	Neural Net Zoo: MLPs, CNNs, RNNs
66	Deep Learning Lecture Review (4)	Sep 5 2024	🟨 DL	CNNs, RNNs, GNNs, Inductive Bias
67	Deep Learning Lecture Review (5)	Sep 10 2024	🟨 DL	Transformers: GELU, LayerNorm, Architecture
68	Deep Learning Lecture Review (6)	Sep 11 2024	🟨 DL	BERT, GPT-2/3/4 & Transfer Learning
69	DL Review – CNNs	Sep 6 2024	🟨 DL	CNN Architecture & Feature Maps
70	DL Review – Transformer	Sep 7 2024	🟨 DL	Self-Attention, QKV, Positional Encoding
71	DL Review – NLP Overview	Sep 8 2024	🟨 DL	NLP Tasks, Goals, Tokenization
72	DL Lecture Review (7)	Sep 10 2024	🟨 DL	Algebra Refresher & ML Math Basics
73	DL Lecture Review (8)	Sep 11 2024	🟨 DL	Foundation Models & Transformer Workflow
74	DL Lecture Review (9)	Sep 12 2024	🟨 DL	Label Noise, Data Leakage, Bias Detection
75–84	Intro to NLP using Deep Learning	Oct 6 2024	🟨 DL	Word Embeddings, CNNs, RNNs, LSTMs, Transformer Overview
85	DL Lecture Review – Lecture 7	Oct 7 2024	🟨 DL	Perplexity, Tokenizers, Embedding Layer
86	DL Lecture Review – Lecture 8 (1)	Oct 8 2024	🟨 DL	Text Generation, Positional Encoding, LoRA
87	DL Lecture Review – Lecture 8 (2)	Oct 9 2024	🟨 DL	GQA, KV Cache, MoE, DPO
88	DL Lecture Review – Lecture 10–12	Oct 10 2024	🟨 DL	Numerical Precision, Hardware & Distributed Training
89	ML Review – Ensemble Method Recap	Oct 11 2024	🟩 ML	Random Forest & XGBoost Revisited

Day	Title / Topic	Date	Category	Key Concepts
90	DL Lecture Review – Background Knowledge	Oct 12 2024	🟨 DL	Softmax Properties, PyTorch Lightning, DataLoader
91	Optimizing Hyperparameters (W&B Integration)	Oct 14 2024	🟨 DL	W&B for Monitoring and Fine-Tuning ResNet-18, Post-Training Evaluations (Dying ReLU, Brightness Robustness)
92	Fine-Tuning Models (1)	Oct 17 2024	🟨 DL	LoRA Understanding, ResNet + SBERT for VQA
93	Fine-Tuning Models (2) & Prompt Engineering	Oct 23 2024	🟨 DL	Embedding Comparison (CLIP vs ResNet + SBERT), Few-Shot Prompt Design
94	Deep Learning Lecture Review – Lecture 13	Oct 30 2024	🟨 DL	LLaMA 3 Architecture (RMSNorm, GQA, RoPE, SwiGLU), Pre-training & Post-training
95	Deep Learning Lecture Review – Lecture 14	Oct 31 2024	🟨 DL	AI Ethics, AI Safety, AGI Challenges
96	Deep Learning Lecture Review – Lecture 15 (1)	Nov 1 2024	🟨 DL	Bias Detection & Mitigation, McNemar's Test, Dataset Bias
97	Deep Learning Lecture Review – Lecture 15 (2)	Nov 2 2024	🟨 DL	Bias Mitigation via Loss Reweighting, Sampling, Adversarial Training, DANN
98	DL Review – Revisiting LLM Models (Tree Map)	Nov 3 2024	🟨 DL	Transfer Learning, NLP Components, Attention Mechanism
99	Deep Learning Lecture Review – Lecture 16	Nov 4 2024	🟨 DL	Uncertainty Estimation, Distribution Shift, Calibration, OOD Detection
100	Deep Learning Lecture Review – Lecture 17 (1)	Nov 12 2024	🟨 DL	Conformal Prediction – Marginal Coverage & Recipes
101	Deep Learning Lecture Review – Lecture 17 (2)	Nov 23 2024	🟨 DL	Adaptive Conformal Prediction, Calibration Set Size Analysis
102	Deep Learning Lecture Review – Lecture 18 (1)	Nov 29 2024	🟨 DL	Data-Centric AI – Crowdsourcing, Annotator Quality, Scaling Laws
103	Deep Learning Lecture Review – Lecture 18 (2)	Dec 9 2024	🟨 DL	Active Learning, SEALS, Dataset Pruning
104	Deep Learning Lecture Review – Lecture 19	Dec 12 2024	🟨 DL	Label Noise, Selection Bias, Data Leakage, Subgroup Error Analysis
105	Deep Learning Lecture Review – Lecture 20 (End)	Dec 18 2024	🟨 DL	Model Drift Detection, Periodic Retraining, Continual Learning
106	Enhancing DL Workflows (Data Distribution)	Dec 19 2024	🟨 DL	Optimization via Data Loading, Profiling, DDP vs DP
107	HW4 – Model Calibration & Conformal Prediction	Dec 20 2024	🟨 DL	Platt Scaling, Label Smoothing, Naïve & Adaptive Prediction Sets
108	HW5 – OOD Detection & Continual Learning	Dec 21 2024	🟨 DL	Maximum Softmax Probability, ODIN, SLDA, IID Streaming

⚙️ 2025 — Starting with MLOps & Statistical Integration

- ***From learning models to building systems:*** *Transitioned from individual model understanding to end-to-end ML systems.*
- *Focused on automation, reliability, and data-centric workflows in production-level MLOps pipelines.*

📊 Phase IV – Applied Statistics (Days 109–154)

> *Goal: Connect classical statistical inference with modern ML applications*

Key Topics:

- Regression (OLS, Interactions, Multicollinearity)
- Classification (Naïve Bayes, LDA, Logistic Regression)
- Unsupervised Learning (PCA, K-Means, GMM, Gower's Distance)
- Model Evaluation & Interpretability

Day	Title / Topic	Date	Category	Key Concepts
109	Exploratory Data Analysis (1)	Jan 25 2025	🟦 STAT	Data Types, Descriptive Statistics, Measures of Location & Variability
110	Exploratory Data Analysis (2)	Jan 26 2025	🟦 STAT	Correlation, Distribution, Visualization Methods
111	Data & Sampling Distributions (1)	Jan 28 2025	🟦 STAT	Sampling, Bias, Central Limit Theorem
112	Data & Sampling Distributions (2)	Jan 29 2025	🟦 STAT	Bootstrap, Confidence Intervals, Normal Distribution
113	Data & Sampling Distributions (3)	Jan 31 2025	🟦 STAT	t-, Chi-Square, F-, Poisson Distributions
114	Statistical Experiments & Significance Testing (1)	Feb 1 2025	🟦 STAT	A/B Testing, Hypothesis Tests, Permutation Tests
115	Statistical Experiments & Significance Testing (2)	Feb 5 2025	🟦 STAT	p-Values, Type I & II Errors, Practical Applications
116	Statistical Experiments & Significance Testing (3)	Feb 6 2025	🟦 STAT	t-Tests, Multiple Testing, False Discovery Rate
117	Statistical Experiments & Significance Testing (4)	Feb 7 2025	🟦 STAT	ANOVA (One- & Two-Way), F-Statistics, Chi-Square
118	Statistical Experiments & Significance Testing (5)	Feb 9 2025	🟦 STAT	Chi-Square Theory, Fisher's Exact Test, Power & Sample Size
119	Regression & Prediction (1)	Feb 11 2025	🟦 STAT	Simple & Multiple Linear Regression, Least Squares
120	Regression & Prediction (2)	Feb 13 2025	🟦 STAT	Model Selection, Cross-Validation, Prediction Using Regression
121	Regression & Prediction (3)	Feb 15 2025	🟦 STAT	Dummy Variables, Factor Variables, Interactions
122	Regression & Prediction (4)	Feb 16 2025	🟦 STAT	Multicollinearity, Confounding Variables, Interpretation
123	Regression & Prediction (5)	Feb 18 2025	🟦 STAT	Outliers, Influential Obs., Heteroskedasticity
124	Regression & Prediction (6)	Feb 19 2025	🟦 STAT	Polynomial Regression, Splines, GAM
125	Revisiting Mathematics Theories (1)	Feb 22 2025	🟦 STAT	Hypothesis Testing, Paired/Independent Samples
126	Revisiting Mathematics Theories (2)	Feb 23 2025	🟦 STAT	Chi-Square & F-Distribution, Inference for Variance
127	Revisiting Mathematics Theories (3)	Feb 24 2025	🟦 STAT	Proportion Inference, Chi-Square Test
128	Revisiting Mathematics Theories (4)	Feb 25 2025	🟦 STAT	Non-Parametric Inference (Wilcoxon Tests)
129	Regression & Prediction (7)	Feb 26 2025	🟦 STAT	Stepwise Regression & Model Selection
130	Regression & Prediction (8)	Feb 28 2025	🟦 STAT	Weighted Regression & Interaction Effects
131	Classification (1)	Mar 1 2025	🟦 STAT	Naïve Bayes – Theoretical Approach & Prediction
132	Classification (2)	Mar 2 2025	🟦 STAT	Discriminant Analysis – Covariance, Default Risk
133	Classification (3)	Mar 3 2025	🟦 STAT	Logistic Regression – Mathematical Foundation (Odds, Logit)

Day	Title / Topic	Date	Category	Key Concepts
134	Classification (4)	Mar 5 2025	🟦 STAT	Logistic Regression – GLM, Fitting, Interpretation
135	Classification (5)	Mar 7 2025	🟦 STAT	Logistic Regression – Model Assessment
136	Classification (6)	Mar 8 2025	🟦 STAT	Evaluating Classifiers – Confusion Matrix, ROC, Lift
137	Classification (7)	Mar 9 2025	🟦 STAT	Handling Imbalanced Data – Over/Undersampling
138	Classification (8)	Mar 14 2025	🟦 STAT	Strategies for Imbalanced Data – SMOTE, Cost-Based Classification
139	Statistical ML (1)	Mar 15 2025	🟦 STAT	KNN – Distance Metrics & One-Hot Encoding
140	Statistical ML (2)	Mar 16 2025	🟦 STAT	KNN – Choosing K, Standardization
141	Statistical ML (3)	Mar 20 2025	🟦 STAT	KNN as Feature Engine + Tree Model Intro
142	Tree Models (2)	Mar 21 2025	🟦 STAT	Recursive Partitioning, Impurity (Gini/Entropy)
143	Tree Models (3)	Mar 25 2025	🟦 STAT	Overfitting Handling, Continuous Prediction
144	Ensemble (1)	Mar 28 2025	🟦 STAT	Bagging & Random Forest – Core Idea
145	Ensemble (2)	Mar 29 2025	🟦 STAT	Random Forest (2) – Variable Importance
146	Boosting (1)	Mar 30 2025	🟦 STAT	AdaBoost & XGBoost Key Concepts
147	Boosting (2)	Mar 31 2025	🟦 STAT	Regularization, Hyperparameters, Cross-Validation
148	Unsupervised Learning (1)	Apr 2 2025	🟦 STAT	PCA – Computing Principal Components
149	Unsupervised Learning (2)	Apr 4 2025	🟦 STAT	PCA – Interpretation & Correspondence Analysis
150	Unsupervised Learning (3)	Apr 6 2025	🟦 STAT	K-Means Clustering – Algorithm & Example
151	Unsupervised Learning (4)	Apr 8 2025	🟦 STAT	K-Means: Determining Optimal K, Interpretation
152	Unsupervised Learning (5)	Apr 13 2025	🟦 STAT	Hierarchical Clustering – Dendrogram, Linkage Methods
153	Unsupervised Learning (6)	Apr 15 2025	🟦 STAT	Model-Based Clustering – Gaussian Mixture Models
154	Unsupervised Learning (7)	Apr 20 2025	🟦 STAT	Scaling, Mixed Data, Gower's Distance

⚙️ Phase V – MLOps Systems (Days 155–203)

> *Goal: Learn scalable ML systems design from data engineering to deployment & monitoring*

Key Topics:

- Data Engineering (ETL, Parquet, Streaming Pipelines)
- Feature Engineering & Data Leakage Prevention
- Experiment Tracking & Model Versioning (W&B)
- Model Serving (Batch vs Online, Cloud vs Edge)
- Monitoring & Drift Detection (Prometheus, Grafana)
- CI/CD Automation (Docker, GitHub Actions, Azure ACI)

Day	Title / Topic	Date	Category	Key Concepts
155	Intro to ML System Design (1)	Apr 24 2025	🟥 MLOps	Business ↔ ML Objectives, Requirements, Iterative Design
156	Intro to ML System Design (2)	Apr 25 2025	🟥 MLOps	Problem Framing, Task Definition, Evaluation Criteria
157	Intro to ML System Design (3)	Apr 26 2025	🟥 MLOps	ML Task Types, Objective Functions
158	Data Engineering Fundamentals (1)	Apr 29 2025	🟥 MLOps	Data Formats (JSON, Parquet), Storage Engines, ETL
159	Data Engineering Fundamentals (2)	May 17 2025	🟥 MLOps	Dataflow Modes – Batch vs Streaming
160	Training Data (1)	May 18 2025	🟥 MLOps	Sampling Methods – Stratified, Weighted, Reservoir
161	Training Data (2)	May 23 2025	🟥 MLOps	Labeling – Hand vs Natural Labels, Active Learning
162	Training Data (3)	May 25 2025	🟥 MLOps	Class Imbalance – Metrics, Sampling, Algorithm-Level Fixes
163	Training Data (4)	May 26 2025	🟥 MLOps	Data Augmentation – Label-Preserving Transformations
164	Feature Engineering (1)	May 29 2025	🟥 MLOps	Handling Missing Values, Scaling, Encoding, Normalization
165	Feature Engineering (2)	May 31 2025	🟥 MLOps	Advanced Features – Positional Embedding, Domain-Specific Engineering
166	Feature Engineering (3)	Jun 2 2025	🟥 MLOps	Data Leakage – Detection & Prevention
167	Model Development & Evaluation (1)	Jun 27 2025	🟥 MLOps	Model Selection, Ensemble (Bagging/Boosting/Stacking)
168	Model Development & Evaluation (2)	Jun 28 2025	🟥 MLOps	Experiment Tracking, Versioning, Distributed Training
169	Model Development & Evaluation (3)	Jun 29 2025	🟥 MLOps	AutoML, NAS, Establishing Baselines
170	Model Development & Evaluation (4)	Jun 30 2025	🟥 MLOps	Offline Evaluation – Perturbation & Invariance Tests
171	Model Deployment & Prediction Service (1)	Jul 1 2025	🟥 MLOps	Batch vs Online Prediction, Unified Pipeline Design
172	Model Deployment & Prediction Service (2)	Jul 2 2025	🟥 MLOps	Model Compression – Low-Rank Factorization, KD, Pruning, Quantization
173	Model Deployment & Prediction Service (3)	Jul 3 2025	🟥 MLOps	Cloud vs Edge Deployment, Model Optimization (AutoTVM, WebAssembly)
174	Data Distribution Shifts & Monitoring (1)	Jul 5 2025	🟥 MLOps	ML System Failures – Feedback Loops & Edge Cases
175	Data Distribution Shifts & Monitoring (2)	Jul 7 2025	🟥 MLOps	Covariate Shift, Label Shift, Concept Drift Detection
176	Data Distribution Shifts & Monitoring (3)	Jul 8 2025	🟥 MLOps	Addressing Drifts – Importance Weighting & Adaptation
177	Data Distribution Shifts & Monitoring (4)	Jul 9 2025	🟥 MLOps	Monitoring vs Observability, ML Metrics & System Reliability
178-190	SQL Fundamentals for Data Analysis (Course Completed)	Jul 11 2025	🟥 MLOps	SQL Basics, Joins, Analytic Functions (Window, Ranking)
191-198	Hands-On Docker (Course Completed)	Aug 11 2025	🟥 MLOps	Docker CLI → Compose → Multi-Container Orchestration
199-200	CI/CD with GitHub Actions + Azure (1)	Sep 2 2025	🟥 MLOps	Build & Push Docker Images → Deploy to Azure ACI via ACR Automation
201–203	CI/CD with GitHub Actions + Azure (2)	Sep 5–8 2025	🟥 MLOps	Full CI/CD workflow with secret management, ACI → AKS scaling

💡 Phase VI – Integration & Review (Day 204–208)

> Theme: Leetcode + DL Review Integration
>
> **Focus:** Strengthening problem-solving and system intuition.

Highlights:

- Python & SQL Problem Solving (Two Sum, Max Subarray, Joins)
- DL Concept Review – CNNs, Transformers, Positional Encoding
- Revisiting Optimizers (SGD, Adam), Autoencoders, Knowledge Distillation
- Connecting Algorithms with Deep Learning Insights

📘 *Core Skills*: Problem Solving • Systems Thinking • Cross-Domain Integration

Day	Title / Topic	Date	Category	Key Concepts
204	DL Review – Optimizers, CNNs & Data Drift	Sep 9 2025	🟨 DL	Optimizers (SGD, Adam), Parameter Sharing in CNNs, Concept Drift Overview
205	Python 175 & SQL Inner Join & DL Review	Sep 12 2025	🟩 ML / 🟨 DL	Two Sum (Python), SQL Joins, Embedding Layers, Autoencoders, Knowledge Distillation
206	Python 217 & SQL 175/176 & DL Review	Sep 13 2025	🟩 ML / 🟨 DL	Contains Duplicate, Second-Highest Salary, Transfer Learning, Fine-Tuning, CNNs
207	Python 53 & SQL 185 & DL Review	Sep 15 2025	🟩 ML / 🟨 DL	Maximum Subarray, Department Top Sales, RNNs, LSTMs, Vanishing/Exploding Gradients
208	Python 121 & SQL 175/176 & DL Review	Sep 16 2025	🟩 ML / 🟨 DL	Best Time to Buy/Sell Stock, Second-Highest Salary, Transformers & Positional Encoding